开始于	2025年06月4日 星期三 15:23
状态	已结束
完成于	2025年06月4日 星期三 15:49
用时	25 分钟 51 秒
评分	**14.00**/21.00 (**66.67**%)

试题 **1**
正确
得分 1.00/1.00 分

In which of the following operations, duplicated columns are eliminated?

选择一项：

- A. Equi-join
- B. Natural join ✔
- C. Condition-join
- D. Cartesian product

试题 **2**

正确

得分 1.00/1.00 分

Building indices on relational tables can ().

选择一项：

- A. improve query processing efficiency. ✔
- B. avoid illegal deletions.
- C. avoid illegal insertions.

试题 **3**

正确

得分 1.00/1.00 分

Generally speaking, in which of the following scenarios a column is not suitable for indexing?

选择一项：

- A. A column that is used in an ORDER BY clause.
- B. A column that has a lot of duplicated values. ✔
- C. A column that is a primary key or foreign key.
- D. A column that is frequently searched.

试题 **4**

正确

得分 1.00/1.00 分

Assume we have two relations R(A, B) and S(B, C, D). There are 2000 tuples in R and 80 tuples in S, and one block can hold 500 R tuples or 20 S tuples. Suppose R and S are already sorted on the attribute B. If we use Sort-Merge algorithm for natural join, at least how many disk blocks need to be read?

选择一项：

- A. 16
- B. 8 ✔
- C. 12
- D. 10

试题 **5**

错误

得分 0.00/1.00 分

Given a set of 10 search key values, we would like to build a B+ tree where the number of pointers for each node is 4. The hight of the tree is no more than ().

选择一项：

- A. 3
- B. 2 ✖
- C. 5
- D. 4

试题 **6**

错误

得分 0.00/2.00 分

Consider the relations r1(A, B, C), r2(C, D, E), and r3(E, F). Assume that there are no primary keys, except the entire schema. Let V (C, r1) = 900, V (C, r2) = 1100, V (E, r2) = 50, and V (E, r3) =100.

Assume that r1 has 1000 tuples, r2 has 1500 tuples, and r3 has 750 tuples. Estimate the size of r1 ⋈ r2 ⋈ r3.

答案： 10227 tuples ✖

试题 **7**

正确

得分 1.00/1.00 分

For large centralized relational databases, the main goal of query optimization is to minimize ().

选择一项：

- A. Memory usage cost
- B. Network communication cost
- C. Computation cost
- D. Disk I/O cost ✔

试题 **8**

错误

得分 0.00/2.00 分

Assume we have two relations R(A, B) and S(B, C, D). There are 2000 tuples in R and 80 tuples in S, and one block can hold 500 R tuples or 20 S tuples. If we use nested-loop algorithm for natural join with R as the outer relation, in the worst case (there is enough memory only to hold one block of each relation), estimate the number of block transfers.

答案: 20 ✖

试题 **9**

正确

得分 1.00/1.00 分

In query processing for relational databases, selection operations should be performed as late as possible.

选择一项:

- ○ 对
- ◉ 错 ✔

试题 **10**

错误

得分 0.00/2.00 分

Consider the relations r1(A, B, C), r2(C, D, E), and r3(E, F), with primary keys A, C, and E, respectively. Assume that r1 has 1000 tuples, r2 has 1500 tuples, and r3 has 750 tuples. Estimate the size of r1 ⋈ r2 ⋈ r3.

答案: 1000 tuples ✖

试题 **11**

正确

得分 1.00/1.00 分

It is generally not possible to build multiple clustering indices on the same relation on different search keys.

选择一项:

- 对 ✔
- 错

试题 **12**

正确

得分 1.00/1.00 分

Suppose a query retrieves only the first K results of an operation and terminates after that. Between demand-driven and producer-driven pipelining, which one would be a better choice for such a query?

选择一项:

- A. Demand-driven ✔
- B. Producer-driven

试题 **13**

正确

得分 1.00/1.00 分

Consider the following table. Dividing salary values into four ranges: S1: below 50,000; S2: 50,000 to below 60,000; S3: 60,000 to below 70,000; and S4: 70,000 and above. Construct a bitmap index on the attribute salary. The bitmap for S3 is ().

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

选择一项：

- A. 100000010000

B. 001000000000

C. 100010010000✔

D. 010101101111

试题 **14**

正确

得分 1.00/1.00 分

In optimization of relation algebra expressions, which of the following operations should be performed as early as possible?

选择一项:

A. Selection✔

B. Join

C. Cartesian product

试题 **15**

正确

得分 1.00/1.00 分

Among the following algorithms that implement the join operation, which one does not involve special access paths or special handling?

选择一项:

A. Nested-loop join✔

B. Indexed nested loop join

C. Merge-join

D. Hash join

试题 **16**

正确

得分 1.00/1.00 分

Building Indices on a table can provide different access paths and speed up query processing.

选择一项:

- 对 ✔
- 错

试题 **17**

正确

得分 1.00/1.00 分

The implementation of Cartesian product is often based on ().

选择一项:

- A. Indexed look join
- B. Sort join
- C. Hash join
- D. Nested loop join ✔

试题 **18**

正确

得分 1.00/1.00 分

The best way to perform the selection operation over a small relation (e.g., that can fit in the memory buffer) is ().

选择一项：

- A. binary search
- B. sort-merge
- C. indexed scan
- D. sequential scan ✔